the plant journal

The Plant Journal (2025) **124**, e70503 doi: 10.1111/tpj.70503

The MdGAMYB-MdHVA22g module confers drought tolerance by mediating γ-aminobutyric acid content and reactive oxygen species scavenging

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Received 25 May 2025; revised 24 August 2025; accepted 13 September 2025.
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SUMMARY

HVA22 is an abscisic acid (ABA)- and stress-induced protein. However, how it is regulated and whether it plays a role under drought stress are largely unclear. In this study, we found that drought-inducible MdHVA22g plays a positive role under drought stress. Overexpression of *MdHVA22g* impairs endoplasmic reticulum (ER) morphology. Further analysis demonstrated that MdHVA22g enhances drought tolerance by upregulating *MdGAD1* and *MdGAD4* expression, thereby leading to increased GAD activity, γ-aminobutyric acid (GABA) accumulation, and enhanced reactive oxygen species (ROS) scavenging. In addition, we identified MdGAMYB as an upstream regulator of *MdHVA22g*. MdGAMYB, the drought-positive regulator, is able to directly bind to the promoter of *MdHVA22g* and activate its expression in response to drought, which results in increased GAD activity, GABA biosynthesis, and ROS detoxification. Therefore, the regulatory cascade of MdGAMYB-MdHVA22g enhances drought tolerance by facilitating ER stress-mediated GABA accumulation and subsequent ROS detoxification in apple. Collectively, our findings reveal a novel regulatory factor of MdHVA22g and elucidate the role of the MdGAMYB-MdHVA22g module in drought tolerance.

Keywords: apple, drought, γ-aminobutyric acid, MdHV22g, MdGAMYB.

INTRODUCTION

Drought is considered as one of the most critical limiting factors on crop quality and yield, particularly as global temperatures continue to rise (Jiang et al., 2022). Drought stress triggers a series of adverse effects in plants, such as damage to cellular membranes (Mahmood et al., 2020), reduced photosynthetic rates (Karami et al., 2025), excessive accumulation of reactive oxygen species (ROS) (Ahluwalia et al., 2021; Samanta et al., 2024), increased ion permeability (Haghpanah et al., 2024), and elevated levels of malondialdehyde (MDA) (Zhang et al., 2021), ultimately leading to metabolic disorder. To cope with drought stress, plants have evolved many strategies. One of the critical intracellular responses to drought is the induction of endoplasmic reticulum (ER) stress, which arises from the accumulation of misfolded or unfolded proteins due to the disruption of ER protein folding capacity (Liu & Howell, 2010). To counteract this, plants activate the unfolded protein response (UPR), a conserved signaling cascade that alleviates ER stress and restores cellular homeostasis (Howell, 2013). However, when ER stress becomes severe or prolonged, it can lead to programmed cell death (Spencer & Finnie, 2020). Recent evidence indicates that ER stress plays a pivotal role in promoting the accumulation of γ-aminobutyric acid (GABA), a non-proteinogenic amino acid implicated in plant stress responses. In *Arabidopsis thaliana*, ER stress markedly upregulates the expression of glutamate decarboxylase (GAD) genes, accompanied by a significant increase in GABA levels, except for *GAD2* (Ozgur et al., 2020). GABA biosynthesis is primarily catalyzed by GAD enzymes, whose activity is regulated by intracellular Ca^{2+} fluctuations, one of the hallmark signals of ER stress (Groenendyk et al., 2021).

GABA is widely present in living organisms (Sarasa et al., 2020), which was first discovered in potato tubers in

1949 (Steward et al., 1949). GABA synthesis and metabolism primarily occur through the GABA shunt pathway by which glutamate undergoes decarboxylation to form GABA under the action of the enzyme glutamate decarboxylase. Subsequently, GABA is converted into succinic semialdehyde through the action of GABA transaminase (GABA-T) (Ramos-Ruiz et al., 2019; Shelp et al., 1999). GABA in plants is involved in various key physiological processes, including the regulation of growth and development at different stages, as well as responses to abiotic stresses. Under drought stress, exogenous GABA enhances drought tolerance in apple plants by increasing biomass, improving leaf relative water content (Cheng et al., 2023), and activating the ABA signaling pathway (Liu et al., 2021). In Arabidopsis, exogenous GABA acts as a signaling molecule that regulates stomatal opening, thereby improving water use efficiency and enhancing drought tolerance (Xu et al., 2021). Beyond its role in stress response, GABA has recently been reported as a significant modulator of ROS in plants. GABA regulates ROS levels through both enzymatic and metabolic mechanisms. Exogenous GABA application has been consistently associated with enhanced activities of key antioxidant enzymes, including superoxide dismutase (SOD), catalase (CAT), ascorbate peroxidase (APX), and peroxidase (POD), thereby promoting ROS detoxification and mitigating oxidative damage (Abd El-Gawad et al., 2021; Feng et al., 2024; Jalil & Ansari, 2020). Additionally, the GABA shunt contributes to maintaining redox balance by bypassing part of the tricarboxylic acid (TCA) cycle, generating NADH and supporting mitochondrial respiration under stress conditions (Ansari et al., 2021; Xu et al., 2024).

HVA22, a TB2/DP1 (deleted in polyposis) protein domain containing protein, was initially identified in the barley aleurone layer cells (Shen et al., 1993; Zhang et al., 2023). HVA22 is distributed across eukaryotes but not in prokaryotes (Shen et al., 2001). In yeast, the homolog of HVA22, Yop1p, is involved in the balanced vesicular trafficking between the ER and the Golgi apparatus (De Antoni et al., 2002). Yop1p is also capable of interacting with the reticulon protein Rt4n/NogoA, thereby regulating protein–protein interactions and modulating ER function (Hu et al., 2008). In plants, the *HVA22* gene family has been identified in several species, including Arabidopsis (Chen et al., 2002), citrus (Gomes Ferreira et al., 2019), tomato (Wai et al., 2022), soybean (Chen et al., 2025), passionfruit (Hou et al., 2024), and cotton (Zhang et al., 2023). Previous studies have shown that HVA22s play crucial roles in regulating plant growth and development. In barley, HVA22 is involved in seed germination and seedling growth by negatively modulating gibberellin (GA)-mediated programmed cell death in aleurone cells (Guo & David Ho, 2008). In Arabidopsis, AtHVA22d is implicated in flower development. Silencing the *AtHVA22d* gene enhances autophagy and leads to a range of developmental abnormalities, including shortened filaments, pistil defects, and impaired pollen function. These defects directly impact seed yield, ultimately resulting in reduced reproductive output (Chen et al., 2009). Moreover, HVA22s display distinct expression and functions in response to various stresses. In Arabidopsis, the gene expression of *AtHVA22s* varies under cold, drought, and salt stress conditions (Chen et al., 2002). Similarly, passionfruit *HVA22* paralogs (*PeHVA22A* and *PeHVA22C*) show drought-induced upregulation but transcriptional suppression under salt and cold stresses (Hou et al., 2024). Functional studies in citrus revealed CcHVA22d mediated drought tolerance through H_2O_2 scavenging mechanisms (Gomes Ferreira et al., 2019). In rice, the *HVA22* homolog gene *OsHLP1* is induced by rice blast disease and enhances the disease resistance of rice by regulating ER homeostasis (Meng et al., 2022). Genomic analyses of cotton species revealed that most *GhHVA22* promoters contain drought-responsive and hormone-related *cis*-elements, with *GhHVA22E1D* specifically conferring drought and salt tolerance through enhanced antioxidant capacity (Zhang et al., 2023).

Apple fruit quality and production are severely affected by drought stress. However, how HVA22s are regulated and whether they are involved in drought tolerance of apple are unclear. In this study, we identified an *HVA22g* gene which is induced by drought stress. MdHVA22g enhances drought tolerance by increasing GABA content and reducing ROS. Furthermore, we revealed that MdGAMYB activates *MdHVA22g* expression and enhances drought tolerance. Taken together, we revealed a novel regulator of MdHVA22g and uncovered the roles of the MdGAMYB–MdHVA22g module in apple drought stress.

RESULTS

Genome-wide identification and analysis of the MdHVA22 family

Based on the sequence similarity, we identified 19 *HVA22* genes in apple, distributed across chromosomes 2, 5, 6, 8–10, and 12–17 (Figure S1a; Table S1). To further investigate the evolutionary relationships of the HVA22s among apple (*Malus domestica*), Arabidopsis (*Arabidopsis thaliana*), rice (*Oryza sativa*), grape (*Vitis vinifera*), and tomato (*Solanum lycopersicum*), we constructed a phylogenetic tree (Figure S1b). The results revealed that the *HVA22* gene family is divided into four subfamilies. The largest subfamily, Group I, contains 31 members, while the smallest subfamily, Group II, has 11 members. Groups III and IV contain 20 and 28 members, respectively (Figure S1b). Apple *HVA22* genes are distributed across all Groups.

Analysis of the *MdHVA22* gene family revealed diverse gene structures and conserved functional motifs. The gene structure analysis of the *HVA22* genes in apple revealed the distribution of coding regions, with the number of exons ranging from 1 (MD02G1239000) to 8 (MD10G1080600) (Figure S2a,b). We then investigated the conserved motifs within the *MdHVA22* gene family and their specific distribution across the *MdHVA22* genes. A total of 10 conserved motifs were identified (Figures S2c and S3), and this analysis showed that all *MdHVA22* genes include Motif 2 (PKTKGAAYVYENFVRPYVAKHETEIDRKL). Furthermore, consistent with the phylogenetic tree analysis, these 10 motifs were classified into four groups: Group I (Motifs 1–4), Group II (only Motif 2), Group III (Motifs 1, 2, 4, and 5), and Group IV (Motifs 2 and 4) (Figure S2a,c). Additionally, conserved domain characterization demonstrated the universal presence of the TB2/DP1 domain across all identified *HVA22s* homologs in apple, Arabidopsis, rice, grape, and tomato (Figures S2d and S4).

Gene duplication analysis in the *HVA22* gene family of apple revealed a total of 11 duplication events in the apple genome (Figure S5), all of which are classified as whole-genome duplications (WGD) or segmental duplications (Table S2). The *Ka*/*Ks* values of the *HVA22* paralogous genes in apple ranged from 0.153 to 0.668 (Table S2). To further understand the evolutionary relationships among *HVA22* members in different plant species, we analyzed syntenic relationships between *MdHVA22* and its homologs from Arabidopsis, rice, grape, and tomato. The results showed that apple is more distantly related to the monocots, as 15 collinear gene pairs were found between apple and rice. In contrast, apple exhibits a closer genetic relationship with other dicots, including Arabidopsis, tomato, and grape, as evidenced by the identification of 21 collinear gene pairs in pairwise comparisons between apple and each of these species (Figure S6; Table S2).

To investigate the response patterns of *MdHVA22* genes to drought stress, we performed qRT-PCR analysis. The results showed that nine *MdHVA22* genes were induced under drought treatment, with *MdHVA22g* being the most significantly upregulated gene. The expression of three *MdHVA22* genes was downregulated, while seven *MdHVA22* genes did not show a response to drought (Figure 1a; Figure S7).

MdHVA22g functions as a positive factor in response to drought stress in apple

Since *MdHVA22g* exhibited the highest induced expression level under drought stress, we selected it for further study. Genetic structure analysis indicated that MdHVA22g contained six exons, and multiple sequence alignment showed that the N-terminal domain exhibits ~80% sequence similarity across different species (Figures S2b and S8). Subcellular localization analysis showed that MdHVA22g was localized in the ER (Figure 1c,d). Tissue-specific expression analysis revealed that *MdHVA22g* was highly expressed in apple leaves, roots, and flowers, but expressed at lower levels in stems (Figure 1b).

To investigate the function of *MdHVA22g* in response to drought stress, we generated transgenic apple plants with both overexpression (OE) and gene interference (RNAi) of *MdHVA22g* in the background of M9-T337 plants (Figure S9a–d). The leaf water loss rate of *MdHVA22g* OE plants was significantly lower than that of M9-T337 plants, while *MdHVA22g* RNAi plants exhibited a significantly higher leaf water loss rate than M9-T337 plants (Figure S10). Then we treated the transgenic *MdHVA22g* OE plants with a 17-day drought treatment, with the well-watered plants as a control. After drought treatment, the survival rate of *MdHVA22g* OE plants (~67%) was higher than that of M9-T337 plants (~43%) (Figure 2a,b). The electrolytic leakage and MDA content of *MdHVA22g* OE plants under drought stress were lower than those of M9-T337 plants (Figure 2c,d). Moreover, *MdHVA22g* OE plants exhibited a higher Fv/Fm (maximum quantum yield of PSII photochemistry) and enhanced photosynthetic capacity than M9-T337 plants (Figure S11a–c). Meanwhile, we treated *MdHVA22g* RNAi plants with a 14-day drought treatment, after which, ~53% of the M9-T337 plants survived, whereas only ~33% of the *MdHVA22g* RNAi plants were still alive (Figure 2e,f). *MdHVA22g* RNAi plants showed higher electrolytic leakage and MDA content (Figure 2g,h), and a lower Fv/Fm and reduced photosynthetic capacity compared with M9-T337 plants (Figure S11d–f). The above results suggest that MdHVA22 is a positive regulator of drought tolerance.

To study whether *MdHVA22g* also has the function of drought resistance in other species, we obtained stably transformed tomato overexpression plants in the background of cultivar 'Alisa Craig' (AC) (Figure S9e,f). Using the *MdHVA22g* OE transgenic tomatoes, we obtained similar results to those in apple. The water loss rate of *MdHVA22g* OE transgenic tomato leaves was lower than that in AC (Figure S12b). After 12 days of drought treatment, ~27% of AC plants survived, while ~55% of *MdHVA22g* OE transgenic tomatoes were still alive (Figure S12a,c). Under drought treatment, *MdHVA22g* OE transgenic tomatoes exhibited lower electrolytic leakage and MDA levels, as well as higher Fv/Fm and photosynthetic rates compared with AC (Figures S12d,e and S13). These results suggest that MdHVA22g could play a positive role in tomato and further confirm the function of MdHVA22g in positively regulating drought tolerance.

MdHVA22g increases plant GABA content by enhancing GAD activity in response to drought

GABA plays a crucial role in regulating plant tolerance to abiotic stresses. Previous studies have shown that ER stress promotes biosynthesis of GABA (Ozgur et al., 2020), a non-protein amino acid playing crucial roles in plant

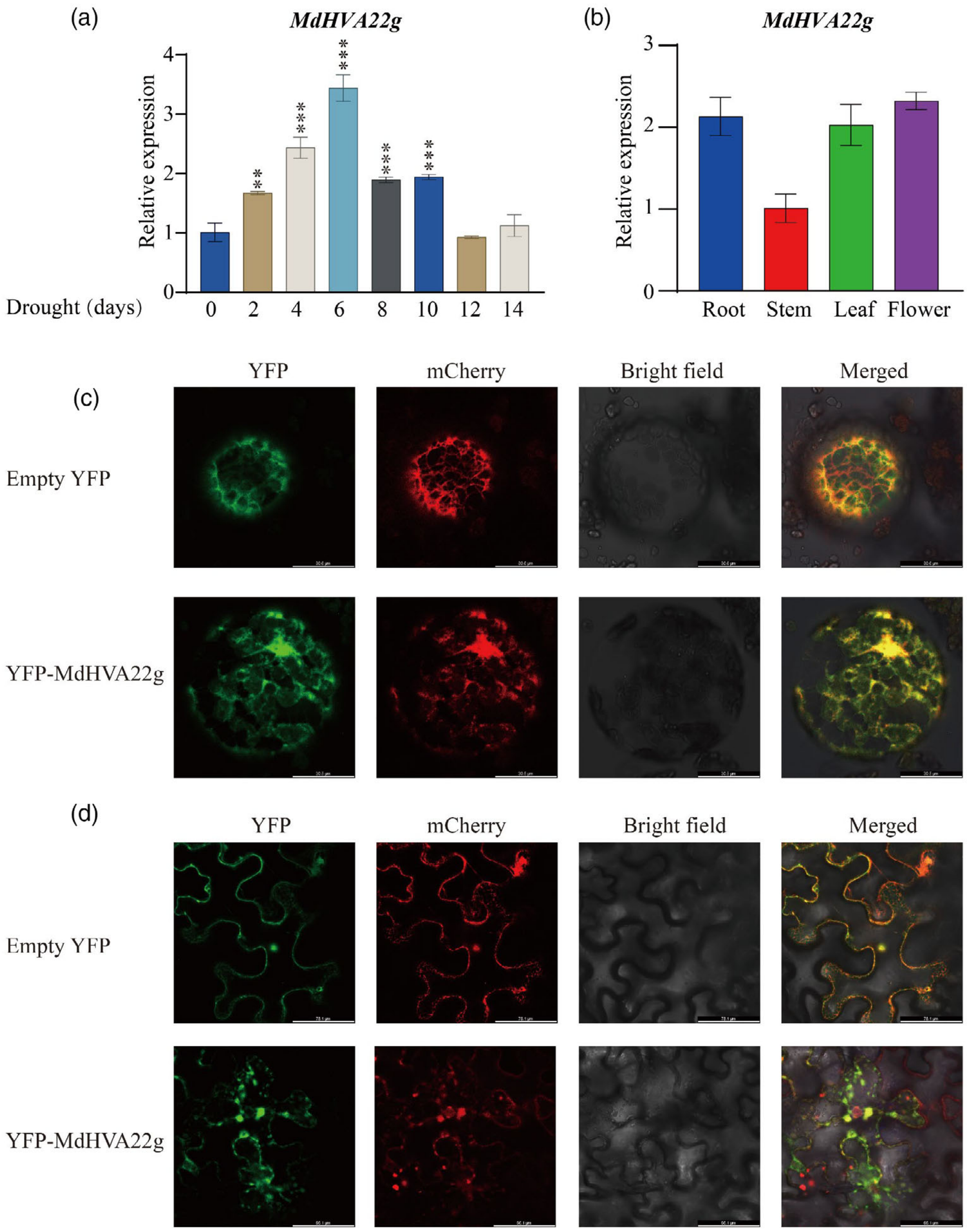

Figure 1. MdHVA22g expression and subcellular localization.
(a) *MdHVA22g* expression under drought treatment in M9-T337 plants. Asterisks indicate significant differences between control and drought treatment.
(b) *MdHVA22g* tissue-specific expression in apple.
(c, d) Subcellular localization of MdHVA22g in *N. benthamiana* protoplasts (c) and leaves (d). mCherry: a red fluorescent protein (RFP) fused with an endoplasmic reticulum marker. Statistical analyses were performed by Student's two-tailed *t*-test (**$P < 0.01$; ***$P < 0.001$). The error bars indicate standard deviations [$n = 3$ in (a, b)].

tolerance to abiotic stresses. As shown in Figure 1, overexpression of *MdHVA22g* induces ER abnormalities (Figure 1c,d). Building on this, we measured the GABA content in M9-T337 and *MdHVA22g* transgenic plants. The results suggested that *MdHVA22g* OE plants exhibited a higher GABA content compared with M9-T337 plants, while *MdHVA22g* RNAi plants had lower GABA levels under both control and drought conditions (Figure 3a,e). Since the GABA shunt pathway is central to GABA biosynthesis in plants, we also assessed GAD activity in these plants under both control and drought conditions. The results were consistent with those of the GABA content

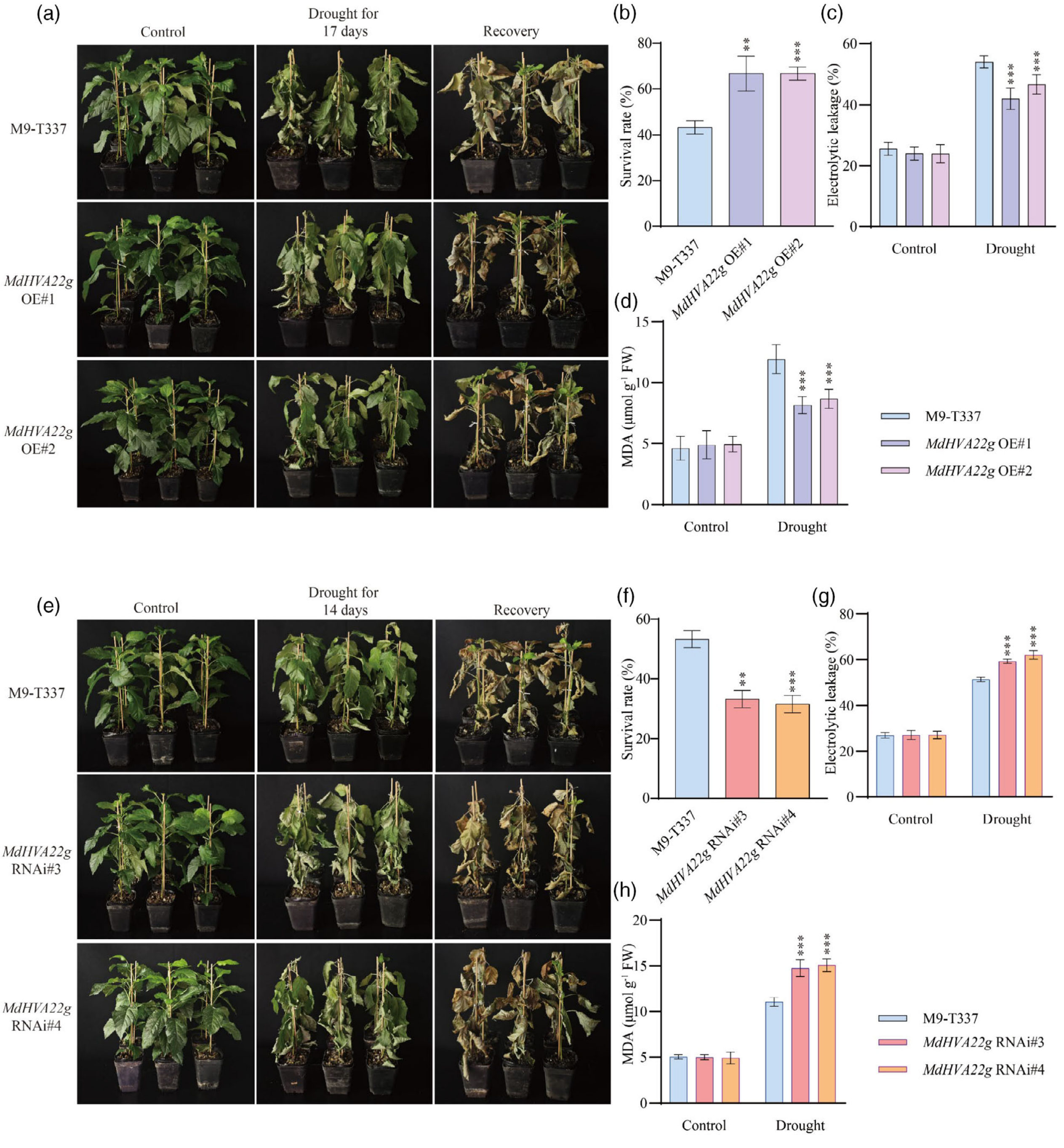

Figure 2. MdHVA22g promotes drought tolerance in apple.
(a) Whole-plant drought tolerance of M9-T337 and *MdHVA22g* OE plants. After drought treatment for 17 days, all plants were transferred to the growth chamber for recovery.
(b) Survival rates of M9-T337 and *MdHVA22g* OE plants.
(c, d) Electrolytic leakage (c) and MDA content (d) of M9-T337 and *MdHVA22g* OE plants under both control and drought conditions.
(e) Whole-plant drought tolerance of M9-T337 and *MdHVA22g* RNAi plants. After 14 days of drought treatment, all plants were transferred to the growth chamber for recovery.
(f) Survival rates of M9-T337 and *MdHVA22g* RNAi plants.
(g, h) Electrolytic leakage (g) and MDA content (h) of M9-T337 and *MdHVA22g* RNAi plants under both control and drought conditions. Asterisks indicate significant differences between M9-T337 and the transgenic lines in each group (control and drought treatment). Statistical analyses were performed by Student's two-tailed *t*-test (** $P < 0.01$; *** $P < 0.001$). The error bars indicate standard deviations [$n = 3$ in (b, f); $n = 8$ in (c, g); $n = 5$ in (d, h)].

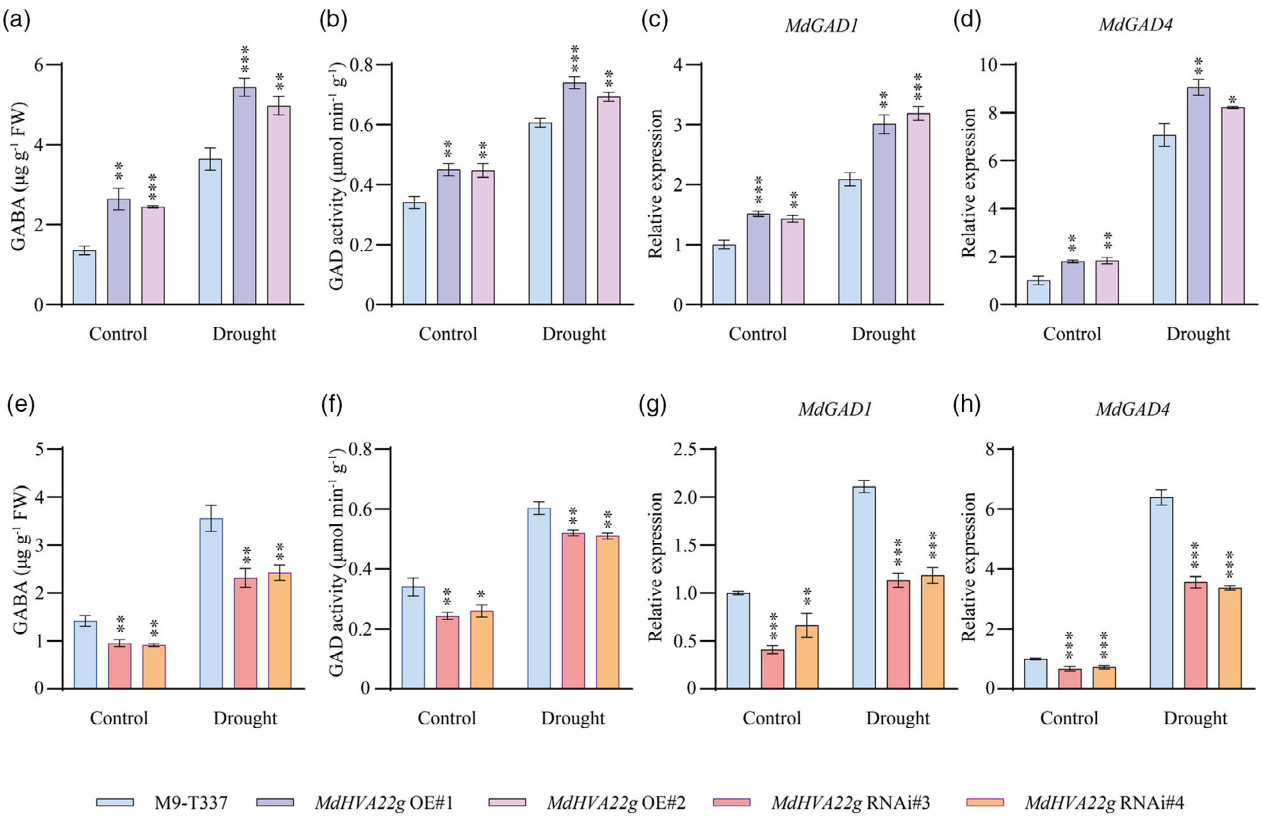

Figure 3. MdHVA22g increases plant GABA content by enhancing GAD activity in response to drought.
(a–d) The GABA content (a), GAD activity (b), *MdGAD1* expression (c), and *MdGAD4* expression (d) of M9-T337 and *MdHVA22g* OE plants under control and drought conditions.
(e–h) The GABA content (e), GAD activity (f), *MdGAD1* expression (g), and *MdGAD4* expression (h) of M9-T337 and *MdHVA22g* RNAi plants under control and drought conditions. Asterisks indicate significant differences between M9-T337 and the transgenic lines in each group (control and drought treatment). Statistical analyses were performed by Student's two-tailed *t*-test (*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$). The error bars indicate standard deviations ($n = 3$).

analysis. Specifically, under both control and drought conditions, *MdHVA22g* OE plants showed higher GAD activity compared with M9-T337 plants, whereas *MdHVA22g* RNAi plants displayed lower GAD activity than M9-T337 plants (Figure 3b,f). Similarly, the GABA content and GAD activity in *MdHVA22g* OE transgenic tomato leaves were higher than those in AC under both control and drought conditions (Figure S12f,g).

To further investigate the molecular mechanism by which *MdHVA22g* regulates GAD activity, we analyzed the expression patterns of genes involved in the GABA shunt pathway in M9-T337 plants and *MdHVA22g* transgenic plants in response to drought treatment. The results revealed that, compared with M9-T337, the expression levels of the GABA biosynthetic genes *MdGAD1* and *MdGAD4* were significantly higher in *MdHVA22g* OE plants under both control and drought conditions (Figure 3c,d). In contrast, the expression levels of the *MdGAD1* and *MdGAD4* were significantly lower in *MdHVA22g* RNAi plants under both control and drought conditions (Figure 3g,h). Additionally, the expression of other GAD genes (*MdGAD2*, *MdGAD3*, *MdGAD5*) and GABA metabolism-related genes (*MdGABA-T1* and *MdGABA-T2*) did not show significant changes (Figure S14). These findings suggest that MdHVA22g may positively regulate GAD activity and GABA content in apple by modulating the expression of *MdGAD1* and *MdGAD4*.

Exogenous GABA enhances the drought tolerance of M9-T337 plants by scavenging ROS

Extensive research has established GABA as a crucial modulator of plant stress responses, demonstrating its capacity to enhance tolerance against various abiotic stressors, including drought, through ROS scavenging mechanisms (Liu et al., 2021; Xu et al., 2024). To further elucidate the ROS regulation of GABA, we conducted a comprehensive investigation using M9-T337 plants subjected to controlled drought conditions with 1 mM exogenous GABA treatment.

Quantitative analysis revealed a ~2.3-fold increase in endogenous GABA levels in treated plants (1 mM GABA application) compared with control plants (0 mM GABA

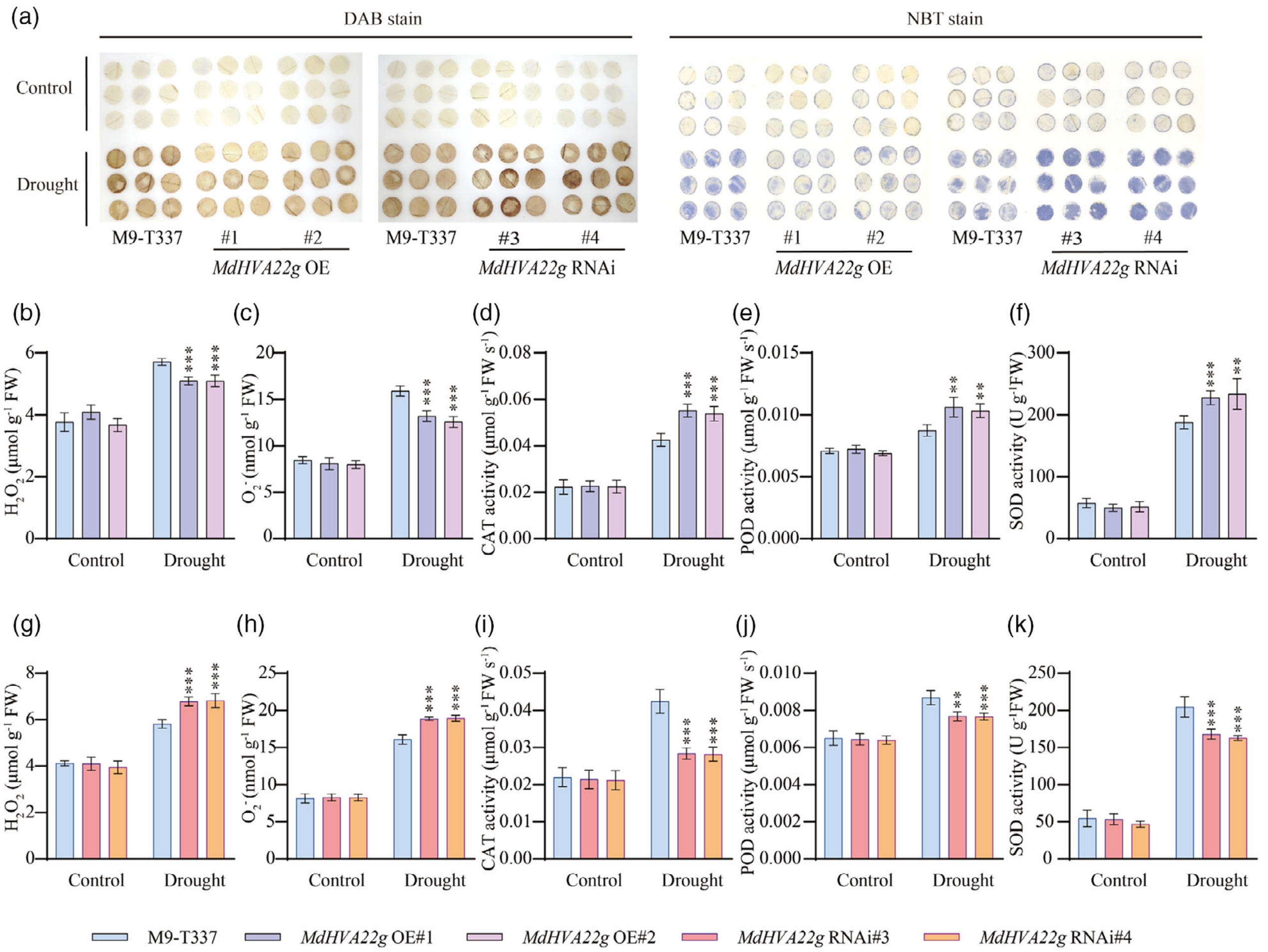

Figure 4. MdHVA22g promotes ROS scavenging under drought stress.
(a) DAB and NBT staining the leaves of M9-T337, *MdHVA22g* OE, and *MdHVA22g* RNAi plants under control and drought conditions.
(b–f) H_2O_2 content (b), O_2^- content (c), CAT activity (d), POD activity (e), and SOD activity (f) of M9-T337 and *MdHVA22g* OE plants under control and drought conditions.
(g–k) H_2O_2 content (g), O_2^- content (h), CAT activity (i), POD activity (j), and SOD activity (k) of M9-T337 and *MdHVA22g* RNAi plants under control and drought conditions. Asterisks indicate significant differences between M9-T337 and the transgenic lines in each group (control and drought treatment). Statistical analyses were performed by Student's two-tailed *t*-test (**$P < 0.01$; ***$P < 0.001$). The error bars indicate standard deviations ($n = 5$).

application) under control conditions, as well as a ~1.8-fold increase under drought conditions (Figure S16b). This enhanced GABA accumulation correlated with significant improvements in the plants during drought stress. Specifically, the GABA-treated plants exhibited a ~15% increased survival rate versus the control plants under drought (Figure S16a,c), accompanied by a ~22% reduction in electrolytic leakage (Figure S16d) and a ~29% decrease in MDA content (Figure S16e). Notably, ROS profiling revealed that the GABA-treated plants accumulated ~25% less hydrogen peroxide (H_2O_2) (Figure S16h) and exhibited ~20% lower levels of superoxide anion $\left(O_2^-\right)$ (Figure S16i) compared with control plants under drought stress. These findings were further corroborated by DAB (3,3′-diaminobenzidine) staining for H_2O_2 visualization and nitroblue tetrazolium (NBT) staining for O_2^- detection, which showed consistent results for both H_2O_2 and O_2^- concentrations (Figure S16f,g). Concurrently, the GABA-treated plants showed enhanced antioxidant enzyme activities under drought, with CAT activity increasing by 38%, POD activity by 24%, and SOD activity by 32% relative to the control plants (Figure S16j–l).

MdHVA22g enhances ROS scavenging in response to drought

To understand the role of MdHVA22 in ROS detoxification in response to drought, we examined ROS accumulation and antioxidant enzyme activities in M9-T337 and *MdHVA22g* transgenic plants under both control and drought conditions. Under drought stress, *MdHVA22g* OE plants demonstrated ~11% lower H_2O_2 concentrations and ~19% reduced O_2^- accumulation (Figure 4b,c). These

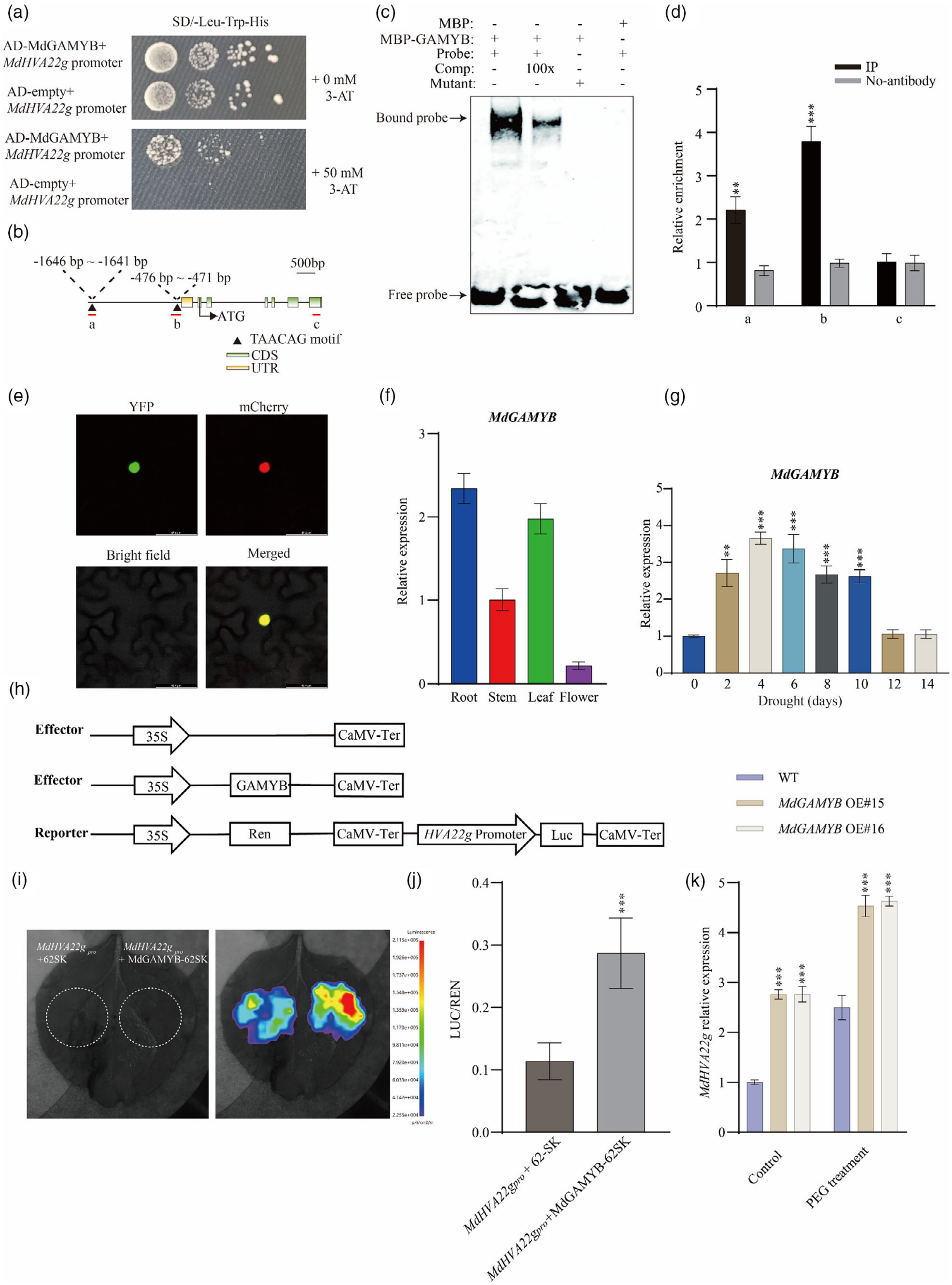
(a)
SD/-Leu-Trp-His
AD-MdGAMYB+ MdHVA22g promoter
AD-empty+ MdHVA22g promoter
+ 0 mM 3-AT
AD-MdGAMYB+ MdHVA22g promoter
AD-empty+ MdHVA22g promoter
+ 50 mM 3-AT
(b)
-1646 bp ~ -1641 bp
-476 bp ~ -471 bp
500bp
ATG
a
b
c
TAACAG motif
CDS
UTR
(c)
MBP:
MBP-GAMYB:
Probe:
Comp:
Mutant:
100x
Bound probe
Free probe
(d)
IP
No-antibody
Relative enrichment
(e)
YFP
mCherry
Bright field
Merged
(f)
MdGAMYB
Relative expression
Root
Stem
Leaf
Flower
(g)
MdGAMYB
Relative expression
Drought (days)
(h)
Effector
35S
CaMV-Ter
Effector
35S
GAMYB
CaMV-Ter
Reporter
35S
Ren
CaMV-Ter
HVA22g Promoter
Luc
CaMV-Ter
WT
MdGAMYB OE#15
MdGAMYB OE#16
(i)
MdHVA22gpro +62SK
MdHVA22gpro + MdGAMYB-62SK
(j)
LUC/REN
MdHVA22gpro+ 62-SK
MdHVA22gpro+MdGAMYB-62SK
(k)
MdHVA22g relative expression
Control
PEG treatment

Figure 5. MdGAMYB directly binds to *MdHVA22g* promoter and positively regulates its expression in response to drought.
(a) Y1H assay demonstrating that MdGAMYB directly binds to *MdHVA22g* promoter. The combinations of pGADT7 + pHIS-$MdHVA22g_{pro}$ were tested for suitable 3-AT concentrations that could inhibit self-activation.
(b) Schematic diagrams of TAACAG motifs in the *MdHVA22g* promoter.
(c) Interaction between MdGAMYB protein and the *MdHVA22g* promoter by EMSA analysis. MBP was used as the negative control. Comp, competitive probe (unlabeled). Mutated probes of *MdHVA22g* promoter fragments had a mutated motif where TAACAG was replaced with AAAAAA.
(d) ChIP-qPCR analysis of MdGAMYB binding to the promoter of *MdHVA22g*. Fragment c and no-antibody serve as negative controls. Asterisks indicate significant differences between fragment c and a or b.
(e) Subcellular localization of MdGAMYB. YFP-MdGAMYB was expressed in *N. benthamiana* leaves. mCherry: a red fluorescent protein (RFP) fused with a nuclear localization signal. Bars = 48.4 μm.
(f) *MdGAMYB* tissue-specific expression in apple.
(g) Expression level of *MdGAMYB* under different durations of drought treatment. Asterisks indicate significant differences between control and drought treatment.
(h) Schematic representation of the reporter and effector.
(i) Relative luciferase activity from the dual-luciferase reporter assays in *N. benthamiana* leaves.
(j) Quantitative analysis of relative luciferase activity from the dual-luciferase reporter assays in *N. benthamiana* leaves. Asterisks indicate significant differences between $MdHVA22g_{\text{pro}}$ + 62sk and $MdHVA22g_{\text{pro}}$ + MdGAMYB-62sk.
(k) Relative expression of *MdHVA22g* in WT and *MdGAMYB* OE calli under control and drought conditions. Asterisks indicate significant differences between WT and the transgenic lines in each group (control and PEG treatment). WT, wild-type. Ren, Renilla luciferase. Luc, firefly luciferase. CaMV-Ter, CaMV terminator. Statistical analyses were performed by Student's two-tailed *t*-test (**$P < 0.01$; ***$P < 0.001$). The error bars indicate standard deviations [$n = 8$ in (j); $n = 3$ in (d, f, g, k)].

reductions correlated with significantly enhanced antioxidant enzyme activities, showing ~1.3-fold higher CAT activity, ~1.2-fold increased POD activity, and ~1.2-fold higher SOD activity compared with M9-T337 plants (Figure 4d–f). Similar results were found in *MdHVA22g* OE transgenic tomatoes (Figure S17). Conversely, *MdHVA22g* RNAi plants exhibited exacerbated oxidative stress under drought, accumulating ~17% more H_2O_2 and ~18% higher O_2^- concentrations than M9-T337 plants (Figure 4g,h), concurrent with ~0.3-fold lower CAT activity, ~0.1-fold decreased POD activity, and ~0.2-fold lower SOD activity compared with M9-T337 plants (Figure 4i–k). Notably, no significant differences in ROS levels or antioxidant enzyme activities between M9-T337 plants and *MdHVA22g* OE or RNAi transgenic plants were observed under control conditions (Figure 4). Consistently, histochemical staining using DAB for H_2O_2 visualization and NBT for O_2^- detection confirmed the ROS accumulation patterns across M9-T337 and *MdHVA22g* transgenic plants under drought stress. *MdHVA22g* OE leaves displayed markedly less DAB and NBT staining compared with the M9-T337 plants, while RNAi plants exhibited intensified coloration, corroborating the quantitative ROS measurements (Figure 4a). The above results demonstrate that MdHVA22g can promote ROS scavenging under drought stress.

MdHVA22g is a direct target of *MdGAMYB*

To identify transcription factors (TFs) regulating *MdHVA22g*, an apple cDNA library was screened using a yeast one-hybrid (Y1H) assay with the *MdHVA22g* promoter fragment as bait. The screening identified *MdGAMYB* (gene ID: MD07G1085000), a MYB-type transcription factor (Figure 5a). Sequence alignment analysis revealed that MdGAMYB shares a high degree of similarity with AtMYB33 (AT5G06100) from Arabidopsis, which belongs to the typical R2R3-MYB subfamily (Figure S18a). In the *MdHVA22g* promoter, we identified two TAACAG motifs which are the potential MdGAMYB binding sites (Figure 5b). To validate the Y1H results, we conducted an electrophoretic mobility shift assay (EMSA). The EMSA results demonstrated that the MBP-MdGAMYB fusion protein could bind to the TAACAG motif in the *MdHVA22g* promoter, and this binding was reduced upon addition of a competing probe and abolished when the binding site was mutated from "TAACAG" to "AAAAAA" (Figure 5c). ChIP-qPCR assays demonstrated that MdGAMYB enriches at the a and b fragments harboring the TAACAG motif within the *MdHVA22g* promoter *in vivo* (Figure 5d). Additionally, MdGAMYB is exclusively located in the nucleus (Figure 5e) and is expressed in roots, stems, leaves, and flowers (Figure 5f).

To further verify the regulatory effect of MdGAMYB on *MdHVA22g*, we performed a dual-luciferase (dual-LUC) assay. The results showed that co-injection of the effector MdGAMYB and the reporter *MdHVA22g* promoter significantly increased LUC gene expression compared with the control (Figure 5h–j). qRT-PCR analysis further confirmed that MdGAMYB positively regulates the expression of *MdHVA22g* under both control conditions and PEG treatment (simulated drought stress) (Figure 5k). These findings collectively suggest that MdGAMYB directly binds to the *MdHVA22g* promoter and positively regulates its expression in response to drought stress.

MdGAMYB positively regulates drought tolerance and GABA content in apple

Considering that MdGAMYB positively regulates the expression of *MdHVA22g*, we hypothesized that MdGAMYB may also play a crucial role in the response to drought stress. *MdGAMYB* expression was induced by

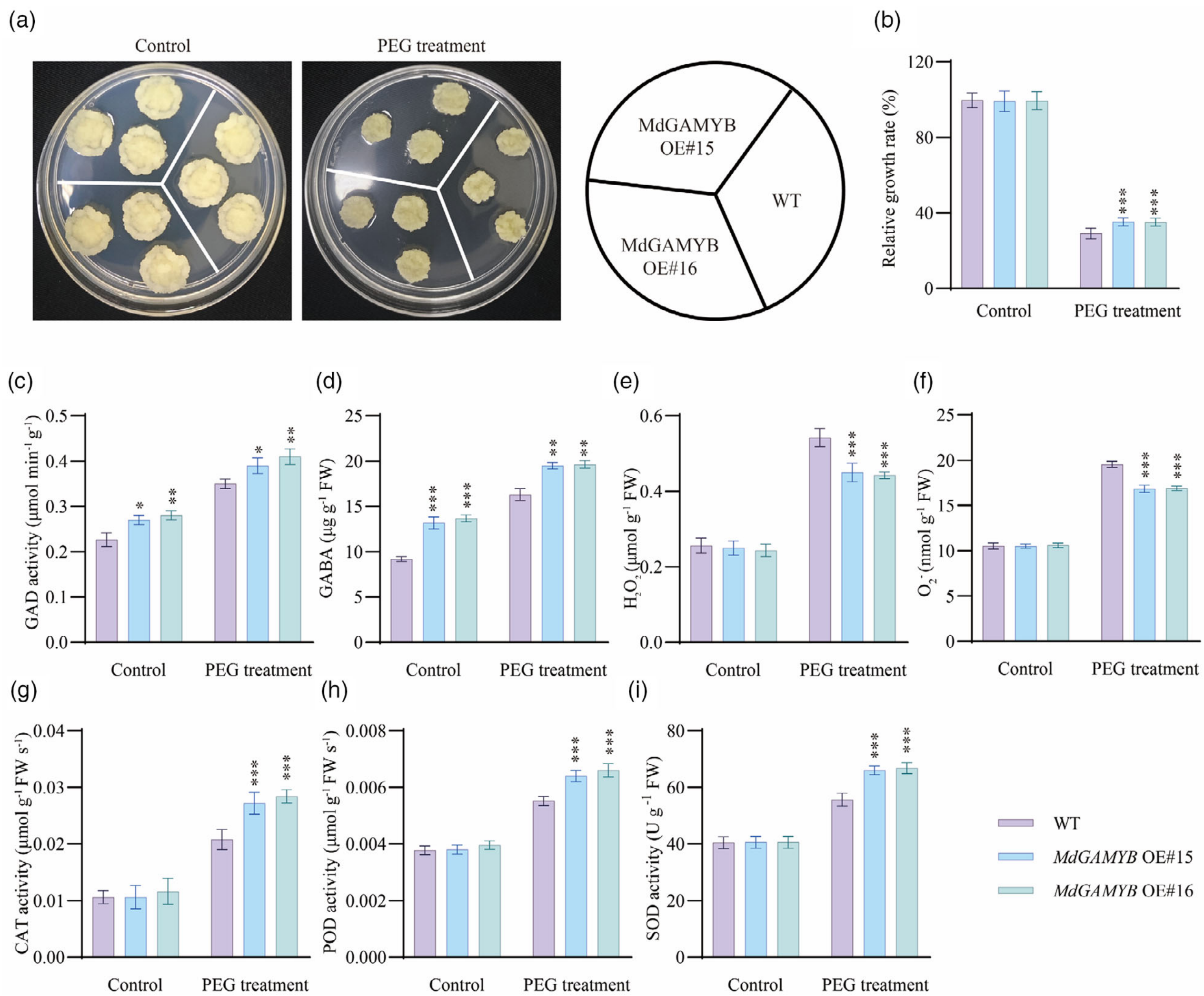

Figure 6. MdGAMYB functions as a positive factor in response to drought stress in apple.
(a–i) Morphology (a), relative growth rate (b), GAD activity (c), GABA content (d), H_2O_2 content (e), O_2^- content (f), CAT activity (g), POD activity (h), and SOD activity (i) of WT and *MdGAMYB* OE calli under control and PEG treatment. WT, wild-type. Asterisks indicate significant differences between WT and the transgenic lines in each group (control and PEG treatment). Statistical analyses were performed by Student's two-tailed *t*-test (*$P < 0.01$; **$P < 0.01$; ***$P < 0.001$). The error bars indicate standard deviations [$n = 9$ in (b); $n = 3$ in (c, d); $n = 8$ in (e–i)].

drought (Figure 5g) with the highest levels observed on the fourth day of drought treatment. To further investigate the role of MdGAMYB in regulating drought tolerance in apple, we generated two *MdGAMYB* OE transgenic calli lines, *MdGAMYB* OE#15 and *MdGAMYB* OE#16, and subjected both wild-type (WT) and *MdGAMYB* transgenic calli to simulated drought stress (Figure S18b–d). The results suggested that under control conditions, the growth rates of *MdGAMYB* OE transgenic calli lines were similar to those of WT. However, under PEG treatment, the *MdGAMYB* OE transgenic calli lines exhibited a significantly higher relative growth rate compared with WT (Figure 6a, b). We then measured the GAD activity and GABA content in both WT and *MdGAMYB* OE lines. The results demonstrated that under both control and PEG treatment conditions, the *MdGAMYB* OE lines showed higher GAD activity and GABA content compared with WT (Figure 6c, d). Additionally, under PEG treatment, the *MdGAMYB* OE lines had significantly lower concentrations of H_2O_2 and O_2^- than WT (Figure 6e,f). Consistently, *MdGAMYB* OE lines exhibited higher activities of CAT, POD, and SOD compared with WT (Figure 6g–i). Taken together, MdGAMYB appears to function as a positive regulator of drought stress tolerance in apple, and this positive regulation might be due to the detoxification of ROS by MdGAMYB.

DISCUSSION

HVA22s play a crucial role in plant growth and development (Chen et al., 2009; Guo & David Ho, 2008) as well as in the response to both biotic and abiotic stresses (Chen et al., 2002; Gomes Ferreira et al., 2019; Hou et al., 2024; Meng et al., 2022; Zhang et al., 2023). However, how it is regulated and whether it plays a role under drought stress are largely unclear. In this study, we identified a module of MdGAMYB-MdHVA22g in apple, which plays a positive role in drought tolerance.

The apple genome contains a total of 19 *HVA22* genes. In comparison, 24, 19, 15, and 13 *HVA22* members were identified in grape, rice, Arabidopsis, and tomato, respectively (Figure S1b). Consistent with previous studies (Wai et al., 2022), the *HVA22* gene family was classified into four distinct subfamilies (Figure S1b). Each subfamily contains *HVA22* genes from various species, indicating that these genes are widely distributed across different species.

Although the number of introns and exons in the *HVA22* genes of apple shows significant variation, all of them contain Motif 2 and the conserved TB2/DP1 domain (Figure S2c,d), suggesting that this motif may be an important structural element involved in the functional activity of *HVA22* genes. Gene duplication is a crucial driving force in plant evolution and a major contributor to genetic diversity (Flagel & Wendel, 2009). The mechanisms of gene duplication include whole genome duplication (WGD), tandem duplication (TD), proximal duplication (PD), transposon-related duplication (TRD), and segmental duplication (DSD) (Li, Yan, et al., 2021; Panchy et al., 2016). We found that all gene duplication events of *HVA22* in apple are associated with WGD (Table S2), suggesting that this is the primary factor responsible for the expansion of the *HVA22* gene family in apple. Moreover, a Ka/Ks ratio of 1 indicates neutral selection, while a ratio <1 suggests purifying selection, and a ratio >1 indicates positive selection (Hurst, 2002). The Ka/Ks ratios for all *MdHVA22* gene duplication events were less than 1, indicating the presence of purifying selection pressure (Table S2). This type of natural selection promotes the removal of deleterious alleles or mutations (Cvijović et al., 2018; Wang et al., 2022), which might be helpful in preserving the conserved functionality of the *MdHVA22* genes in apple. In addition, the expression pattern of *MdHVA22* genes in response to drought was diverse, highlighting their functional diversity.

Under abiotic stress, GABA rapidly accumulates in plants and plays a critical role in mediating stress responses (Islam et al., 2024; Jin et al., 2023; Mishra et al., 2023; Suhel et al., 2023; Yuan et al., 2023). The alleviation of stress sensitivity by GABA should be attributed to mitigated ROS accumulation. In Arabidopsis, the accumulation of GABA in the *gaba-t*/*pop2* mutant enhances plant tolerance to salt and hypoxia stress by facilitating ROS scavenging (Su et al., 2019; Wu et al., 2021). Similarly, in tomato, SlGAD2 improves cold tolerance by promoting GABA accumulation, which in turn enhances antioxidant enzyme activity and anthocyanin content (Wang, Zhang, Wang, Khan, et al., 2024). Additionally, SlGAD1 contributes to salt–alkali tolerance by increasing GABA levels, thereby promoting ROS scavenging and regulating the ethylene signaling pathway (Wang, Zhang, Wang, Ma, et al., 2024). Our results showed that under drought conditions, 1 mM GABA treatment reduced H_2O_2 and O_2^- levels while significantly increasing survival rate and the activities of antioxidant enzymes (CAT, POD, and SOD) (Figure S16). Furthermore, under drought stress, *MdHVA22g* OE plants exhibited significantly higher activities of antioxidant enzymes than M9-T337 plants, while their H_2O_2 and O_2^- levels were significantly lower than those in M9-T337 plants (Figure 4). Given the high similarity of the N-terminal domain of HVA22 between apple and tomato, we further investigated the gene function in tomato. Similarly, in *MdHVA22g* OE transgenic tomato plants, the antioxidant enzyme activities and ROS levels were consistent with those observed in apple (Figure S17). Collectively, these findings suggest that MdHVA22g enhances drought tolerance by increasing GABA content, which in turn facilitates ROS scavenging.

ER is a key organelle for protein synthesis in plants (Eichmann & Schäfer, 2012). Under stress conditions, ER homeostasis can be disrupted, resulting in enhanced sensitivity to ER stress (Meng et al., 2022). To mitigate ER stress, the transcription factors bZIP17 and bZIP28 translocate to the nucleus, where they activate the expression of UPR genes (Liu et al., 2007; Liu & Howell, 2010). HVA22, a homolog of yeast Yop1p, is involved in the formation of the tubular ER network and vesicle trafficking (Xiang et al., 2023). In rice, overexpression of the *HVA22* family protein OsHLP1 disrupts the tubular ER network structure and induces the expression of disease-responsive genes, thereby enhancing resistance to rice blast disease (Meng et al., 2022). Similarly, in this study, MdHVA22g was localized in the ER in plant cells, and its overexpression in tobacco resulted in abnormal ER morphology (Figure 1c,d). Under both control and drought conditions, the expression levels of UPR-related genes *MdIRE1*, *MdbZIP17*, and *MdbZIP60* genes were significantly increased in *MdHVA22g* OE transgenic plants. In contrast, the expression of these genes in *MdHVA22g* RNAi transgenic plants showed the opposite trend. These data indicated that MdHVA22g is involved in the ER stress response (Figure S15). Furthermore, we observed that overexpression of *MdHVA22g* under drought conditions was associated with increased GABA levels and reduced ROS accumulation (Figures 3 and 4). Given that ER stress is known to promote GABA biosynthesis (Ozgur et al., 2020), MdHVA22g-induced ER morphology disruption may enhance drought tolerance by stimulating GABA production and facilitating ROS scavenging.

In summary, we identified the module of MdGAMYB-MdHVA22g in apple and elucidated its molecular role in apple drought response. We found that the MdGAMYB-

MdHVA22g regulatory module promotes drought stress tolerance by regulating GABA levels and ROS detoxification. Our study provides insights into understanding the molecular mechanisms of apple in response to drought.

MATERIALS AND METHODS

Identification of *HVA22s* and phylogenetic tree construction

Full-length HVA22 amino acid sequences from different species were submitted for phylogenetic tree analysis. Sequence alignment was performed using MUSCLE (http://www.ebi.ac.uk/Tools/msa/muscle). The phylogenetic tree was constructed using MEGA11 (Tamura et al., 2021), with the neighbor-joining method for statistical analysis. The phylogeny was tested using the bootstrap method with 1000 replications, and the substitution model used was the Poisson model.

Structural and syntenic analysis of *HVA22s*

TBtools (Chen et al., 2020) was used to visualize the UTR, exon, and intron gene structures of the *MdHVA22* genes. The specific conserved motifs of the *MdHVA22* genes were analyzed using the MEME online tool (https://meme-suite.org/meme/tools/meme), with a maximum of 10 motifs set. MCScanX software was used for synteny analysis as previously described (Ma et al., 2023). Additionally, the Ka and Ks substitution rates for each syntenic gene pair were calculated using the Ka/Ks Calculator.

Plant materials and growth conditions

M9-T337 (wild-type) and transgenic plants were cultured on MS medium (4.43 g L^{-1} MS salts, 30 g L^{-1} sucrose, 0.1 mg L^{-1} 6-BA, 0.2 mg L^{-1} IAA, and 7.5 g L^{-1} agar) at pH 5.8–6.0 under long-day conditions (14 h light and 10 h dark, 25°C) for rooting and stable transformation. After 1 month of growth, they were transferred to ½ MS medium for rooting, which included 2.22 g L^{-1} MS salts, 30 g L^{-1} sucrose, 0.5 mg L^{-1} IBA, 0.5 mg L^{-1} IAA, and 7.5 g L agar at pH 5.8–6.0 under the same growth conditions for an additional month. Finally, the plants were transplanted into soil and grown in a growth chamber (16 h light and 8 h dark, 25°C) at Northwest A&F University, Yangling, China (34°20′ N, 108°24′ E) for 2 months.

Apple calli (*Malus domestica* cv. Orin) were cultured on MS medium containing 4.43 g L^{-1} MS salts, 30 g L^{-1} sucrose, 0.4 mg L^{-1} 6-BA, 1.5 mg L^{-1} 2,4-D, and 7.5 g L^{-1} agar at pH 5.8–6.0. The cultures were maintained at 25°C in the dark.

The tomato (*Solanum lycopersicum* cv. Alisa Craig, AC) seeds were sterilized and then cultured on ½ MS medium (2.22 g L^{-1} MS salt, 30 g L^{-1} sucrose) under long-day conditions (14 h light and 10 h dark, 25°C). After 7 days, the germinated seedlings were ready for soil transplantation or could be used as material for genetic transformation.

Vector construction and genetic transformation

'Golden Delicious' apple leaves were used for gene cloning of *MdHVA22g* and *MdGAMYB*. The coding sequences (CDS) of *MdHVA22g* and *MdGAMYB* were cloned into the pK2GW7 vector and pGWB414 to create overexpression plasmids. To silence *MdHVA22g*, a 214-bp fragment of *MdHVA22g* was inserted into the pK7WIWG2D vector. The resulting plasmids were then introduced into *Agrobacterium tumefaciens* EHA105 and GV3101 for transformation.

Water loss assay

Two-month-old M9-T337 and *MdHVA22g* transgenic plants grown in soil were selected for the study. Water loss assay was performed as described previously (Jiang et al., 2022). The fourth to sixth mature leaves, with similar sizes, counting from the apex, were detached for the water loss experiment. Leaf weights were recorded at 0, 20, 40, 60, 80, 140, 200, 320, 440, and 560 min after the initiation of natural water loss. Thirteen biological replicates were used.

The leaf water loss assay for AC and *MdHVA22g* OE transgenic tomatoes followed the same procedure as described above, with slight modifications to the water loss time points. Leaf weights were recorded at 0, 60, 120, 180, 240, 300, 360, 420, and 480 min after water loss, and the leaf water loss rate was calculated. Ten biological replicates were used.

Drought stress treatment

For the drought stress treatment in apple, 2-month-old M9-T337 and *MdHVA22g* transgenic plants, grown under consistent soil conditions, were subjected to drought stress. Prior to treatment, the plants were irrigated to achieve soil saturation, and the soil volumetric water content (VWC) was measured using time domain reflectometry (TDR). The plants were then allowed to undergo natural water loss until the VWC reached 0. Leaf wilting was documented through photography, and after 14 days of rehydration, the plant survival rate was assessed. Twenty biological replicates were used for each treatment.

For the drought stress treatment in tomato, 1-month-old AC and *MdHVA22g* OE plants, grown under consistent soil conditions, were subjected to drought stress. Similarly, the plants were irrigated to soil saturation before treatment, and their weights were recorded for quantification. The plants were then allowed to undergo natural water loss until the VWC reached 0 after 12 days. Leaf wilting was also documented through photography, and after 7 days of rehydration, the plant survival rate was evaluated. Twenty biological replicates were used for each treatment.

Exogenous GABA treatment

The exogenous GABA treatment was conducted as described previously (Cheng et al., 2023). M9-T337 plants were transplanted into a growth chamber for an additional 2 months. Subsequently, the plants were irrigated with 0 or 1 mM GABA for 30 days prior to the short-term drought treatment.

Polyethylene glycol (PEG) treatment

PEG treatment was performed as described previously (Niu et al., 2022). WT and *MdGAMYB* OE calli, subcultured on MS medium for 2 weeks, were used as experimental materials. Equal weights of both WT and *MdGAMYB* OE calli were placed on either the control group [MS medium containing 1.2 g L^{-1} 2-(*N*-

morpholino) ethane sulfonic acid (MES)] or the PEG treatment (simulated drought stress) group (MS medium containing 1.2 g L^{-1} MES and pretreated with PEG8000 [−0.7 MPa]). The calli were incubated at 25°C under dark conditions for 15 days. The fresh weight of the calli was measured, and the relative growth rate of the calli was subsequently calculated. Nine biological replicates were used for each treatment.

Relative electrolytic leakage

After 10 days of drought treatment, the M9-T337 and *MdHVA22g* transgenic plants were sampled by collecting the fourth to sixth leaves from the apex. The samples were collected from eight different plants, with three leaves collected from each plant as one biological replicate. Eight biological replicates were used for each treatment. Using a 5 mm hole punch, leaf discs were prepared and submerged in 8 mL ddH_2O for 4 h. The electrical conductivity of ddH_2O (R0) and the sample conductivity (R1) were measured. The samples were then placed in boiling water at 100°C for 30 min, allowed to cool to room temperature, and the conductivity (R2) was measured. Finally, relative electrolytic leakage (REL) was calculated using the formula: $\mathrm{REL} = [(R1 - R0)/(R2 - \mathrm{R0})] \times 100\%$.

Antioxidant enzyme activity assay and MDA assay

After 10 days of drought treatment, the fourth to sixth leaves from the apex of M9-T337 and *MdHVA22g* transgenic plants were collected for antioxidant enzyme activity assay and MDA assay. After 14 days of PEG treatment, the calli of WT and *MdGAMYB* OE were collected for antioxidant enzyme activity assay and MDA assay. POD, CAT, and MDA were determined according to the methods described previously (Niu et al., 2022). SOD was determined using detection kits (Suzhou Comin Biotechnology, Suzhou, China).

Determination of H_2O_2 and O_2^- content

After 10 days of drought treatment, the fourth to sixth leaves from the apex of M9-T337 and *MdHVA22g* transgenic plants were harvested to measure H_2O_2 and O_2^- content. After 14 days of PEG treatment, the calli of WT and MdGAMYB OE were collected to measure H_2O_2 and O_2^- content according to the methods described previously (Li, Hou, et al., 2021).

GABA content and GAD activity assay

Fresh plant leaves and calli were ground in liquid nitrogen, and 0.1 g of the powdered sample was placed in a 2 ml centrifuge tube. To this, 1 ml of amino acid extraction solution (50% ethanol with 0.1 M HCl) was added, and the mixture was thoroughly vortexed before being sonicated for 20 min. The sample was then centrifuged at 13 523 ***g*** for 20 min at 4°C, and the supernatant was collected for further analysis. The supernatant was filtered through a 0.22 μm organic filter membrane and subsequently diluted 20-fold with chromatographic-grade acetonitrile for liquid chromatography-mass spectrometry (LC–MS) analysis (LC: AC, ExionLC; MS: Q-trap5500, AB Sciex Pret. Ltd., Framingham, MA, USA). Mobile phases A and B were 0.1% methanoic acid and acetonitrile, respectively, and the flow rate was 0.3 ml min^{-1}. The 5 μl of each sample was injected for measurement. GAD activity was determined using detection kits (Suzhou Comin Biotechnology, Suzhou, China).

Electrophoretic mobility shift assay (EMSA)

The *MdGAMYB* gene was cloned into the pMAL-c5X vector, and the resulting plasmids were transformed into *E. coli* strain Rosetta2 (DE3). Expression of the recombinant protein was induced for 12 h at 16°C. The MBP-MdGAMYB fusion protein was purified using amylose resin (NEB, Ipswich, MA, USA). EMSA analysis was performed using the LightShift™ Chemiluminescent EMSA Kit (Thermo Scientific, Waltham, MA, USA). Biotin-labeled primers and oligonucleotide probes used in the assays are listed in the Table S3.

Dual-luciferase reporter assay (dual-LUC)

The *MdGAMYB* gene was cloned into the pGreenII62-SK vector (*35S::MdGAMYB*), while the promoter fragments of *MdHVA22g* were cloned into the pGreenII-0800 vector (*ProMdHVA22g::LUC*). The resulting plasmids were introduced into *Agrobacterium tumefaciens* GV3101, and *Agrobacterium*-mediated transformation was used for the transient transformation of tobacco leaves. Samples were harvested for analysis 3 days post-transformation. Each treatment group consisted of eight biological replicates. Firefly luciferase and Renilla luciferase activities were measured using the Dual-Luciferase Reporter Assay System (Yeasen, Shanghai, China). The primer sequences are listed in the Table S3.

RNA extraction and qRT-PCR analysis

Total RNA was extracted from plant leaves and calli tissues using the CTAB method, with each sample comprising three biological replicates. Residual genomic DNA was eliminated using DNase I (Thermo Scientific, Waltham, MA, USA). The DNA-free RNA was subsequently reverse-transcribed into cDNA using the RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific). qRT-PCR analysis was conducted using the ChamQ Universal SYBR qPCR Master Mix (Vazyme, Nanjing, China) on a Bio-Rad CFX96™ system. *MdMDH* was used as the reference gene. Gene-specific primers for qRT-PCR are listed in Table S3.

Chlorophyll fluorescence and photosynthesis measurement

The leaves of apple and tomato plants were kept in the dark for 20 min before being measured. Chlorophyll fluorescence parameters were determined using a HEXAGON-IMAGING-PAM (Heinz Walz, Germany).

The photosynthetic rate was measured using a LI-COR 6400 (USA) after 10 days of drought treatment between 9:00 and 11:00 a.m. Leaves were measured using a LED red and blue light source with 1000 μmol m^{-2} sec^{-1} light intensity and 500 μmol sec^{-1} air flow. Five plants from both the control and drought groups were randomly selected for measurement. For consistency, the fourth to sixth leaves from the apex were chosen, ensuring that the selected leaves had similar size and developmental stage. Five biological replicates were used for each treatment.

Yeast one-hybrid assay (Y1H)

The promoter fragment of *MdHVA22* was cloned into the pHIS vector to generate the bait plasmid. This bait plasmid was then transformed into yeast strain Y187, which was subsequently co-transformed with an apple cDNA library. The yeast cells were plated onto selective media containing 3-amino-1,2,4-triazole (3AT) to suppress the activity of endogenous histidine biosynthesis. Positive colonies, which were capable of growing on selective agar plates, were identified, indicating a potential interaction between the bait and library proteins. These positive clones were further analyzed to identify the interacting proteins.

The CDS of *MdGAMYB* was cloned into the pGADT7 vector for yeast one-hybrid analysis. This recombinant plasmid was then transformed into yeast strain Y187, which contained the bait plasmid. The transformed yeast cells were plated onto selective SD/−Leu/−Trp/−His medium containing 3-AT to screen for positive DNA–protein interactions. The primer sequences are listed in the Table S3.

Subcellular localization of MdHVA22g and MdGAMYB

The CDS sequences of *MdHVA22g* and *MdGAMYB* were cloned into the pEarleyGate104 vector to generate fused expression constructs of YFP-MdHVA22g and YFP-MdGAMYB. These constructs were then introduced into *A. tumefaciens* strain GV3101, followed by selection and identification of positive colonies. *Agrobacterium*-mediated transformation was carried out by infiltrating 4-week-old tobacco leaves. Samples were collected and analyzed 3 days post-transformation. The NLS-RFP and ER-RFP vectors were used as controls for nuclear and ER localization, respectively. YFP and RFP signals were detected using a Nikon A1R/A1 confocal microscope (Nikon, Tokyo, Japan).

ChIP-qPCR analysis

ChIP-qPCR analysis of apple calli was performed with minor modifications to the method described previously (Song et al., 2024). *MdGAMYB* OE transgenic calli pretreated with PEG8000 were cross-linked with cross-linking buffer (0.4 M sucrose, 10 mM Tris–HCl, pH 8.0, 1 mM EDTA, and 1% formaldehyde). Immunoprecipitation was performed either in the presence or absence of the anti-MYC antibody, followed by precipitation with ChIP-grade Protein A/G agarose beads (Thermo Scientific). Relative enrichment of promoter fragments of *MdHVA22g* was detected by qRT-PCR analysis. Three biological replicates were used. Primers are listed in Table S3.

Statistical analysis

Statistical analysis was carried out by Student's two-tailed *t*-test. Asterisks indicate significant differences ($*P < 0.05$; $**P < 0.01$; $***P < 0.001$).

AUTHOR CONTRIBUTIONS

QG, CN, and PC conceived and designed the research. PC, XY, JZ, XX, JH, FM, YZ, and YL performed the experiments. PC, XY, FM, and DZ analyzed the data. All authors read and approved this manuscript.

ACKNOWLEDGMENTS

We thank Dr. Chunxiang You from Shandong Agricultural University for providing wild-type "Orin" calli and the high performance computing (HPC) platform of Northwest A&F University (NWAFU) for providing computing resources. We are grateful to Dr. Jing Zhang (Horticulture Science Research Center, Northwest A&F University, Yangling, China) for providing professional technical assistance with LC–MS/MS analysis. This work was supported by grants from the Shenzhen Science and Technology Program (JCYJ20220530161402006), Science and Technology Program of Xinjiang Production and Construction Corps (2024DA010), National Natural Science Foundation of China (32202459), and the Start-up Funds of Northwest A&F University to Chundong Niu (2452024117).

CONFLICT OF INTEREST

The authors declare no conflict of interest and have no relevant financial or non-financial interests to disclose.

DATA AVAILABILITY STATEMENT

The data supporting the results can be found in this paper and its supplementary files.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Chromosome distribution and phylogenetic analysis of *MdHVA22* genes.

Figure S2. Structural analysis of *MdHVA22* genes in apple.

Figure S3. MEME analysis of conserved motifs in MdHVA22 proteins.

Figure S4. Conserved domain analysis of MdHVA22 proteins in Arabidopsis (a), rice (b), tomato (c), and grape (d).

Figure S5. Collinearity analysis of the *MdHVA22* genes in apple genome.

Figure S6. Syntenic analyses of *MdHVA22* genes between apple and four other four species.

Figure S7. Gene expression of the *MdHVA22* gene family under drought treatment in M9-T337 plants.

Figure S8. Sequence alignment of HVA22 proteins in rice, Arabidopsis, apple, tomato, and grape.

Figure S9. Identification of the *MdHVA22g* transgenic plants.

Figure S10. Water loss rate of M9-T337 and *MdHVA22g* transgenic plant leaves.

Figure S11. Fv/Fm fluorescence and net photosynthetic rate of M9-T337 and *MdHVA22g* transgenic plants under control and drought conditions.

Figure S12. MdHVA22g promotes drought tolerance in tomato.

Figure S13. Fv/Fm fluorescence and net photosynthetic rate of AC and *MdHVA22g* transgenic plants under control and drought conditions.

Figure S14. Expression of genes related to GABA biosynthesis and metabolism in M9-T337 and *MdHVA22* transgenic plants.

Figure S15. Expression of genes related to unfolded protein response (UPR) in M9-T337 and *MdHVA22* transgenic plants.

Figure S16. Exogenous GABA enhances drought tolerance of M9-T337 plants by scavenging ROS.

Figure S17. MdHVA22g promotes ROS detoxification in tomato under drought.

Figure S18. Identification of the *MdGAMYB* transgenic calli.

Table S1. Identification and sequence information of *HVA22* gene family in *Malus domestica*.

Table S2. Collinearity analysis of the *HVA22* gene family.

Table S3. List of primers used in this study.

Table S4. Gene IDs of GABA biosynthesis and metabolic pathways.

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